

Toronto Opioid Overdose Dashboard and Analytical Report

Analysis of opioid-related paramedic responses, geographic patterns, healthcare utilization, harm reduction, deaths, and drug-checking findings in Toronto

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Executive Summary

This report presents a Power BI dashboard integrating multiple indicators of opioid-related harms and responses in Toronto. The dashboard combines paramedic-response data, healthcare utilization, opioid-toxicity deaths, neighbourhood patterns, supervised consumption service activity, naloxone distribution, opioid agonist therapy dispensing, and drug-checking findings. The project was developed as a professional analytics case study demonstrating data preparation, data modelling, DAX development, geographic integration, dashboard design, and responsible interpretation of public-health information.

Suspected opioid-overdose calls attended by Toronto Paramedic Services peaked in 2021 and generally declined afterward. Calls decreased by approximately 33% from 2024 to 2025, while average monthly fatal calls also declined. Monthly patterns varied considerably across years, and the analysis did not identify a consistent seasonal pattern.

The 2025 geographic analysis showed that paramedic-attended incidents were concentrated primarily in downtown Toronto. Moss Park recorded both the highest number of calls and the highest population-adjusted call rate. Differences between neighbourhood rankings by count and rate demonstrated the importance of examining both service volume and relative population burden.

Opioid-poisoning emergency department visits and hospitalizations were lower from January to June 2025 than during the same months in 2023 and 2024. Adults aged 25-44 and males recorded the highest numbers of healthcare utilization. Opioid-toxicity deaths also declined from the higher levels observed in 2023, although the burden remained substantial. Adults aged 25-44 recorded the highest number of deaths, private dwellings were the most common living arrangement, and private residences were the most frequently reported incident location.

Harm-reduction and treatment indicators followed different trends. Supervised consumption service visits declined after 2022, although comparisons were affected by site closures, different operating periods, and partial 2025 data. Naloxone-kit distribution increased sharply and remained high through 2025, while the number of individuals dispensed opioid agonist therapy declined gradually.

Drug-checking findings highlighted continued unpredictability in samples submitted as expected fentanyl. Expected-opioids accounted for an increasing share of checked samples, while the proportion fully meeting expectations declined. Benzodiazepine-related detections decreased in

recent years, whereas veterinary tranquilizers increased substantially. Fluorofentanyl and methylfentanyl-related detections peaked in 2024 before declining in 2025.

Overall, several recent indicators moved downward, but the findings should not be interpreted as evidence that opioid-related harm has declined uniformly or that a single intervention caused the changes. The datasets measure different events and cover different reporting periods. The dashboard is therefore best used to identify broad trends, geographic and demographic patterns, changes in service activity, and emerging features of the unregulated drug supply.

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Introduction

Opioid poisoning remains a complex public-health concern in Toronto, affecting emergency-service demand, healthcare utilization, mortality, harm-reduction services, treatment systems, and the composition of the unregulated drug supply. No single dataset captures the full scale or circumstances of opioid-related harm. A broader understanding therefore requires examining several complementary indicators while recognizing that they represent different events, populations, and reporting systems.

This project presents an interactive Power BI dashboard that integrates publicly available data from Toronto Public Health, Public Health Ontario, the Ontario Drug Policy Research Network, Toronto's Drug Checking Service, and other government and administrative sources. The analysis covers suspected opioid-overdose calls attended by Toronto Paramedic Services, neighbourhood-level geographic patterns, opioid-poisoning emergency department visits and hospitalizations, opioid-toxicity deaths, supervised consumption service activity, naloxone-kit distribution, opioid agonist therapy dispensing, and drug-checking findings.

The purpose of the dashboard is to provide a high-level view of how opioid-related harms and responses have changed over time, where recorded paramedic incidents were concentrated, which demographic groups had the highest healthcare utilization and mortality counts, and how harm-reduction, treatment, and drug-supply indicators evolved. The project is intended for public-health planners, analysts, community organizations, harm-reduction and treatment providers, policy professionals, and others seeking an accessible overview of the available Toronto data.

The analysis is descriptive and observational. The indicators are examined separately and are not combined into a single estimate of overdose incidents or affected individuals. Differences in case definitions, reporting periods, data availability, and geographic coverage are considered throughout the report, and external research is used only to provide context rather than establish the causes of Toronto-specific trends.

Project Objectives

Overall objective

This project aims to provide a comprehensive, high-level assessment of opioid-related harms and responses in Toronto by integrating indicators from emergency services, healthcare utilization, mortality records, harm-reduction and treatment programs, and drug-checking services.

The dashboard examines the following areas:

- suspected opioid-overdose calls attended by Toronto Paramedic Services, including fatal and non-fatal responses;
- geographic patterns in paramedic-attended calls across Toronto neighbourhoods;
- opioid-poisoning emergency department visits and hospitalizations;
- opioid toxicity death trends and selected demographic and circumstantial characteristics;
- supervised consumption services, naloxone distribution, and opioid agonist therapy;
- drug-checking samples submitted as expected fentanyl, including whether samples met expectations and which additional substances were detected.

The analysis is intended to identify trends, population groups, geographic areas, and changes in the unregulated drug supply that may help support public-health planning and harm-reduction decision-making.

Intended Audience

The dashboard and analytical report are designed for users who need a high-level view of opioid-related harms and service responses in Toronto, including:

- public-health planners and analysts;
- harm-reduction and treatment organizations;
- community health and social-service agencies;
- municipal and provincial policy analysts;
- healthcare and emergency-service organizations;
- researchers and private or non-profit organizations working in substance-use health and opioid-toxicity prevention.

Analytical Questions

1. How have opioid-related paramedic responses, healthcare utilization, and deaths changed over time in Toronto?
2. Are there recurring monthly patterns in suspected opioid-overdose paramedic calls?
3. Which neighbourhoods and population groups experience the highest recorded burden?
4. What do supervised consumption service activity, naloxone distribution, and opioid agonist therapy indicate about harm-reduction and treatment responses?
5. What substances are detected in samples submitted as expected fentanyl, and how often do the results meet expectations?
6. What data limitations must be considered when interpreting and comparing these indicators?

Data Sources and Coverage

This analysis integrates data from multiple public-health surveillance systems, administrative databases, reports, and interactive data tools to provide a broad view of opioid-related harms and responses in Toronto. The combined indicators cover paramedic-attended suspected opioid overdoses, emergency department visits and hospitalizations, opioid toxicity deaths, harm-reduction and treatment services, and drug-checking findings. Because these indicators come from different data systems and reporting periods, they are analyzed as separate measures and are not combined to estimate a total number of overdose incidents.

Indicator	Primary or original data source	Period covered	Data accessed through
Suspected opioid-overdose paramedic calls	Toronto Paramedic Services, Electronic Patient Care Record	August 2017 to May 2026	Toronto Public Health, Toronto Overdose Information System
Geographic distribution of paramedic-attended	Toronto Paramedic Services, Electronic Patient Care Record	January to December 2025	Toronto Public Health, Calls to Paramedic Services for Suspected Opioid

suspected opioid-overdose calls by neighbourhood			Overdoses: Geographic Information
Opioid-poisoning emergency department visits and hospitalizations - monthly trends	National Ambulatory Care Reporting System for emergency department visits; Discharge Abstract Database for hospitalizations; Ontario Ministry of Health, IntelliHealth Ontario	January 2017 to June 2025	Toronto Public Health, Toronto Overdose Information System
Opioid-poisoning emergency department visits and hospitalizations by age group and sex	National Ambulatory Care Reporting System for emergency department visits; Discharge Abstract Database for hospitalizations; Ontario Ministry of Health, IntelliHealth Ontario	2015 to 2024	Public Health Ontario, Substance Use and Harms Tool
Opioid toxicity deaths - monthly trend	Public Health Ontario, Interactive Opioid Tool, for January 2015 to December 2017; Office of the Chief Coroner for Ontario, Coroner's Opioid Investigative Aid, for January 2018 onward	January 2015 to January 2026	Toronto Public Health, Toronto Overdose Information System
Opioid toxicity deaths by age group, living arrangement,	Office of the Chief Coroner for Ontario	2020 to 2024	Toronto Public Health, Annual Summary of Opioid

and incident location			Toxicity Deaths in Toronto - 2024
Supervised consumption service visits, overdose events, and naloxone administration	NEO database and The Works, Toronto Public Health	August 2017 to March 2025	Toronto Public Health, Toronto Overdose Information System
Naloxone kits distributed	Ontario Naloxone Program for Pharmacies and Ontario Naloxone Program	January 2017 to March 2026	Ontario Drug Policy Research Network, Ontario Opioid Indicator Tool - Harm Reduction
Opioid agonist therapy	Ontario administrative health and prescription-drug databases	January 2013 to December 2025	Ontario Drug Policy Research Network, Ontario Opioid Indicator Tool - Opioid Agonist Therapy
Drug-checking samples and substances detected	Toronto's Drug Checking Service	2020 to 2025	Toronto Public Health, Toronto Overdose Information System

Data Preparation and Dashboard Development

The datasets used in this project were obtained from multiple public-health surveillance systems, administrative databases, published reports, and interactive data tools. Because the source files differed in structure, reporting frequency, date format, geographic coverage, and indicator definitions, the data required cleaning, transformation, and standardization before they could be analyzed together in Power BI.

Data Preparation

Data preparation was completed using Microsoft Excel, Power Query, and Power BI. The main preparation activities included:

- reviewing source documentation and technical notes to understand the definition, scope, and limitations of each indicator;
- restructuring downloaded files and published tables into analysis-ready formats;
- removing fields that were not required for the final analysis;
- standardizing column names, data types, category labels, and reporting periods;
- converting separate year and month fields into consistent date values;
- creating month names, abbreviated month labels, month numbers, quarter fields, and standardized time-period columns;
- separating indicators such as emergency department visits and hospitalizations when they were included in the same source file;
- identifying missing, suppressed, preliminary, and partial-year data;
- standardizing demographic categories, including age group and sex;
- validating transformed values against the original reports, dashboards, and downloadable source files.

Monthly datasets were assigned a standardized **Time Period** field representing the first day of each reporting month. This allowed datasets from different sources to be connected consistently to a central Date table and filtered by year and month.

Annual datasets, such as healthcare utilization by age group and sex and selected opioid-toxicity death characteristics, were connected using a separate year field because monthly dates were not available.

Counts, percentages, rates, and proportions were retained as separate measures. For example, total paramedic-call volume was not treated as equivalent to the population-adjusted call rate, and supervised consumption service visits were not treated as the number of unique clients.

Data Modelling

A relational data model was developed in Power BI to integrate the different subject areas while preserving the original level of detail and unit of measurement of each dataset.

The model contains separate tables for:

- suspected opioid-overdose paramedic responses;
- emergency department visits and hospitalizations;
- opioid-toxicity deaths;
- supervised consumption service activity;
- naloxone-kit distribution;
- opioid agonist therapy;
- neighbourhood-level paramedic calls and population;
- healthcare utilization by age group and sex;
- death characteristics, including age group, living arrangement, and incident location; and
- drug-checking samples and substances detected.

A central **Date Table** was connected to datasets containing monthly records. A separate **Year Table** was used for annual datasets that did not contain month-level information. These dimension tables supported consistent time filtering while respecting the reporting frequency of each source.

A disconnected **Healthcare Utilization Selector** table was also created to allow users to switch selected visuals between emergency department visits and hospitalizations. DAX measures return the appropriate indicator based on the user's selection.

The datasets were not combined into a single fact table because they represent different events, populations, reporting systems, and time periods. A paramedic response, emergency department visit, hospitalization, supervised consumption service overdose, naloxone kit, OAT dispensing record, drug-checking sample, and opioid-toxicity death should not be interpreted as equivalent records or added together to estimate a total number of overdose incidents.

Several datasets were imported or created during the exploratory and development stages but were not used in the final dashboard. These tables were excluded from the final analytical findings.

A visual overview of the final Power BI data model is presented in *Appendix A, Figure A1*.

DAX Measures and Analytical Calculations

DAX measures were developed to support the dashboard's KPI cards, percentages, rankings, dynamic labels, and interactive visuals.

Key calculations included:

- total suspected opioid-overdose paramedic calls;
- fatal and non-fatal paramedic-attended calls;
- fatal and non-fatal call percentages;
- monthly and annual paramedic-call totals;
- monthly rankings within each year;
- total emergency department visits and hospitalizations;
- highest healthcare-utilization age group and sex;
- total opioid toxicity deaths;
- age group with the highest number of deaths;
- most frequently reported living arrangement and incident location;
- supervised consumption service visits and overdose events;
- percentage of service visits involving an overdose;
- percentage of overdose events involving naloxone administration;
- naloxone kits distributed;
- individuals dispensed opioid agonist therapy;
- expected-opioid and expected-fentanyl sample proportions;
- percentages of samples containing selected noteworthy substances;
- neighbourhood paramedic-call totals;
- neighbourhood call rates per 100,000 population;

- neighbourhood rankings by call volume and population-adjusted rate.

Measures were also designed to return blank values when the selected reporting period was unavailable in a particular dataset. This prevented missing data from being displayed as zero or from producing a leading category based on an unrelated year.

Dynamic measures and field selectors were used where appropriate so that one visual could display either emergency department visits or hospitalizations without duplicating the entire chart.

Geographic Data Integration

The geographic analysis used 2025 neighbourhood-level suspected opioid-overdose paramedic-call data and the City of Toronto's 158-neighbourhood geographic boundary file.

Neighbourhood names and identifiers were reviewed and standardized before the call data were linked to the geographic boundaries. Census population data were also incorporated to support the calculation of call rates per 100,000 population.

The geographic page presents two related measures:

- total paramedic calls by neighbourhood; and
- paramedic calls per 100,000 population.

These measures were kept separate because they answer different analytical questions. Total call volume shows where Toronto Paramedic Services attended the greatest number of incidents, while the population-adjusted rate provides a relative comparison among neighbourhoods of different population sizes.

A choropleth map, ranked tables, and KPI cards were used to identify neighbourhoods with the highest call counts and rates. Areas with no reported call record in the source dataset were displayed separately from neighbourhoods with low call values.

Dashboard Development

The final Power BI report contains seven interactive pages.

1. KPI Overview

The overview page presents high-level indicators from the main areas of the project, including paramedic responses, geographic hotspots, healthcare utilization, opioid toxicity deaths, and supervised consumption services.

Year and month slicers allow users to explore available reporting periods. A note informs users that some indicators are not available for every selected month or year.

2. Paramedic Response Trends

This page analyzes changes in suspected opioid-overdose paramedic calls over time. It includes:

- total paramedic calls;
- fatal and non-fatal call rates;
- monthly call volumes;
- annual call trends;
- monthly fatal calls and fatal-call percentages;
- a year-by-month heatmap.

The heatmap ranks months within each year to help identify recurring periods of relatively high or low call volume without allowing years with larger totals to dominate the colour scale.

3. Geographic Hotspots

The geographic page compares Toronto neighbourhoods using total paramedic-call volume and population-adjusted call rates. It includes a choropleth map, top-ten neighbourhood rankings, and KPI cards for the neighbourhood with the highest count and highest rate.

4. Healthcare Utilization

This page examines opioid-poisoning emergency department visits and hospitalizations. It includes:

- total ED visits and hospitalizations;
- monthly trends;
- a month-by-year heatmap;
- healthcare utilization by age group;
- healthcare utilization by sex;
- KPI cards identifying the leading demographic categories.

A selector allows users to switch relevant visuals between emergency department visits and hospitalizations.

5. Deaths Due to Opioid Poisoning

The deaths page combines monthly mortality trends with annual demographic and circumstantial information. It presents:

- total opioid toxicity deaths;
- deaths by age group;
- living arrangements;
- incident locations;
- monthly trends;
- KPI cards identifying the leading categories.

Monthly death data and detailed annual characteristics cover different periods. This distinction is identified in the dashboard and considered when interpreting the results.

6. Harm Reduction and Treatment

This page presents supervised consumption service activity, naloxone distribution, and opioid agonist therapy. It includes:

- supervised consumption service visits by location and year;
- overdose events recorded during service visits;
- overdose events involving naloxone administration;
- overdose events as a percentage of visits;
- naloxone kits distributed by provider type;
- individuals dispensed opioid agonist therapy.

These indicators are displayed together to provide a broad view of harm-reduction and treatment activity. However, changes in one indicator are not interpreted as directly causing changes in another without further analysis.

7. Drug-Checking Results

The drug-checking page focuses on samples submitted as expected fentanyl or other opioids. It includes:

- the proportion of checked samples submitted as expected opioids;

- the proportion submitted as expected fentanyl;
- the percentage in which expected fentanyl was detected;
- noteworthy substances found in checked samples;
- selected high-potency opioids and fentanyl analogues;
- yearly trends in benzodiazepines, veterinary tranquilizers, fluorofentanyl, and methylfentanyl.

The KPI year selector applies to selected cards, while the trend visuals display the complete available period to show changes in the unregulated drug supply over time.

Visual Design and Navigation

A consistent visual design was applied throughout the dashboard to improve usability and create a recognizable project identity. The design includes:

- a blue-and-white colour theme;
- orange emphasis for primary KPI values;
- consistent circular icons and KPI-card layouts;
- a fixed top navigation menu;
- page-specific titles and subtitles;
- left-side filter panels;
- standardized source notes;
- consistent visual containers, spacing, and alignment.

Charts were selected according to the purpose of each indicator. Line charts were used for time trends, heatmaps for recurring monthly patterns, choropleth maps for neighbourhood comparisons, ranked tables for high-burden areas, and bar or column charts for demographic and categorical comparisons.

Dynamic titles, labels, and filter-responsive measures were used to help users understand the reporting period and indicator currently displayed.

Data Validation and Quality Checks

Dashboard outputs were reviewed against the original source reports, interactive tools, and downloaded files after data transformation and measure development.

The quality checks included:

- comparing totals with published source values;
- confirming chronological month sorting;
- checking table relationships and filter behaviour;
- validating neighbourhood, age-group, sex, and death-category rankings;
- confirming the calculation of rates, percentages, and denominators;
- reviewing missing and incomplete reporting periods;
- checking that unavailable years returned blank values where appropriate;
- confirming that partial-year data were not presented as complete annual totals;
- reviewing source labels and period coverage on each page.

Where a recent year contained only partial data, comparisons were limited to equivalent reporting periods when appropriate, or the limitation was clearly identified. The dashboard preserves the distinction among data systems and does not combine different indicators into a single estimate of total overdose cases.

Dashboard analysis and key findings

Paramedic Response Trends

Figure 1.

Paramedic Response Trends dashboard page with 2023–2025 selected.



Note. KPI cards and selected-year charts reflect the active year filters. The annual trend and seasonal heatmap display the broader available reporting period.

Paramedic-call data do not represent the number of people who use opioids in Toronto or the total number of opioid-poisoning incidents. They capture suspected opioid-overdose events attended by Toronto Paramedic Services and therefore exclude incidents in which paramedics were not contacted or involved.

Annual paramedic-attended calls reached their highest level in 2021. Call volume declined sharply in 2022, increased temporarily in 2023, and then resumed its downward trend. Total calls decreased from 4,302 in 2024 to 2,873 in 2025, representing a decline of approximately **33%**. The monthly chart also shows substantial variation, with calls falling noticeably during the first half of 2025 before increasing toward the end of the year.

The heatmap of complete calendar years does not show a consistent seasonal pattern across the reporting period. Relatively high call volumes occurred from March to May in some years, including 2019, 2022, and 2024. In other years, the highest monthly volumes occurred later in the year. For example, several of the highest values in 2020, 2021, and 2025 were recorded during the fall or winter months. Therefore, no single month or season consistently recorded the highest call volume across all years.

Fatal paramedic-attended calls also fluctuated from month to month but showed an overall decline. The average monthly number of fatal calls decreased from approximately 19 in 2024 to 11 in 2025, a reduction of about **42%** based on the rounded monthly averages. However, the fatal-call percentage did not decline uniformly because it is influenced by changes in both fatal calls and total attended calls. Fatal counts and fatal-call percentages should therefore be interpreted together.

Because these data are observational and capture only incidents involving Toronto Paramedic Services, the analysis cannot determine what caused the decline. Changes may reflect several factors, including actual changes in overdose events, whether 911 was called, service use, harm-reduction activity, changes in the unregulated drug supply, or other conditions not measured by this dataset.

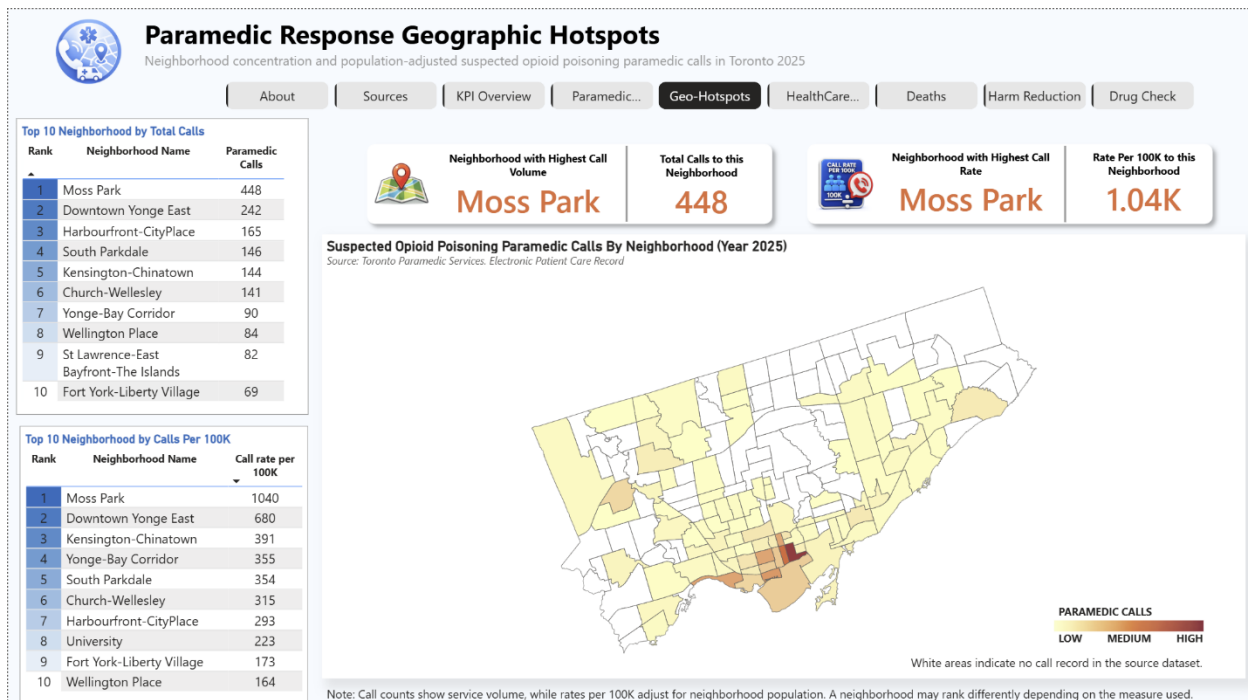
Key finding

Suspected opioid-overdose calls attended by Toronto Paramedic Services have generally declined since their 2021 peak, including a 33% reduction from 2024 to 2025. However, monthly patterns varied across years, and the data do not demonstrate a consistent seasonal pattern or explain the causes of the decline.

Geographic Hotspots of Paramedic-Attended Calls

Figure 2.

Paramedic Response Geographic Hotspots dashboard page for 2025.



Note. Call counts represent the number of suspected opioid-overdose incidents attended by Toronto Paramedic Services, while rates per 100,000 population adjust for neighbourhood population. White areas indicate that no call record was reported for the neighbourhood in the source dataset.

The geographic analysis shows that suspected opioid-overdose paramedic calls were concentrated primarily in and around downtown Toronto. Moss Park recorded the highest call volume in 2025, with **448 attended calls**, substantially more than Downtown Yonge East, which ranked second with 242 calls.

Moss Park also recorded the highest population-adjusted call rate, at approximately **1,040 calls per 100,000 population**. Downtown Yonge East ranked second at 680 calls per 100,000, followed by Kensington-Chinatown at 391 and Yonge-Bay Corridor at 355.

Although the highest-ranked neighbourhoods were generally concentrated in the central city, the rankings differed somewhat depending on whether total call volume or population-adjusted rate was used. Harbourfront-CityPlace ranked third by total calls but seventh by rate, while

University appeared in the top ten by rate but not by total call volume. This demonstrates that neighbourhood population affects how geographic burden is interpreted.

The choropleth map reinforces the concentration of higher call activity in downtown and nearby neighbourhoods. However, the map identifies where paramedics encountered patients, not necessarily where the person lived or where the overdose began. The findings should therefore be interpreted as patterns in attended incident locations rather than residential risk.

Neighbourhood rates should also be interpreted cautiously. Rates based on relatively small populations can appear high even when the absolute number of calls is lower. For this reason, call counts and population-adjusted rates are presented together rather than using either measure alone.

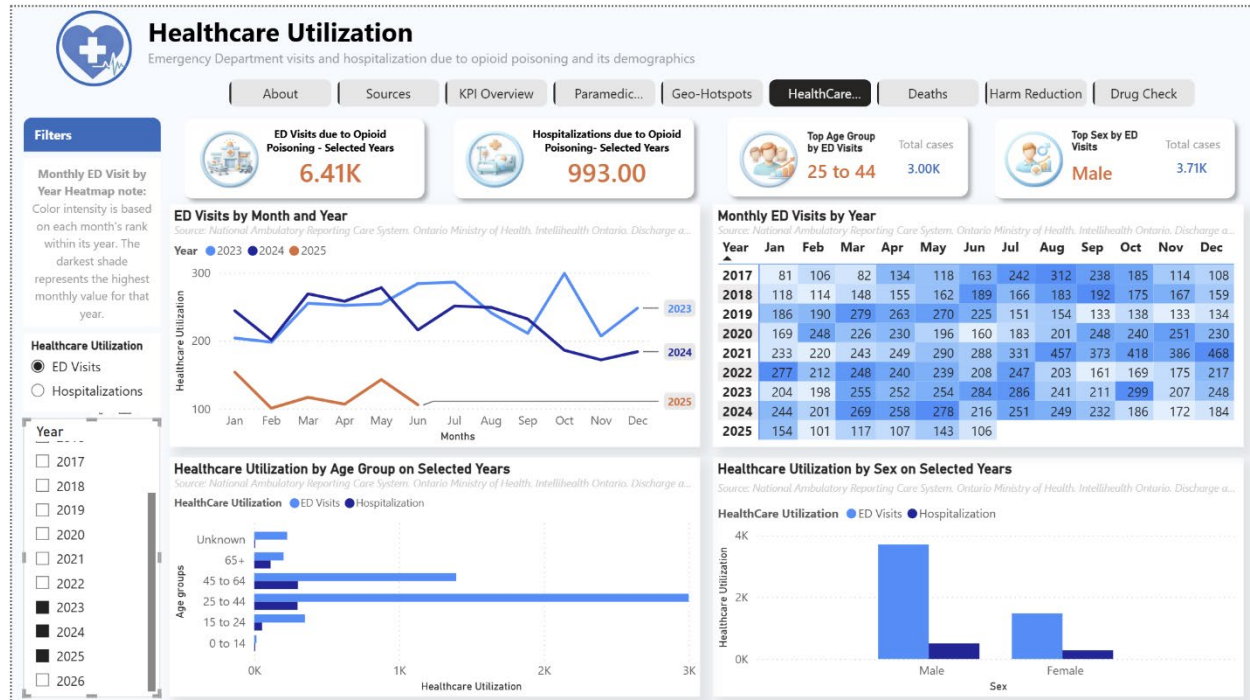
Key finding

Moss Park had both the highest number and the highest population-adjusted rate of suspected opioid-overdose paramedic calls in 2025. More broadly, the greatest recorded burden was concentrated in downtown Toronto, although some neighbourhoods ranked differently when population size was taken into account.

Healthcare Utilization

Figure 3.

Healthcare Utilization dashboard page with 2023–2025 selected.



Note. Monthly 2025 data are available only from January to June. Comparisons with earlier years should therefore use the same January-to-June period rather than complete annual totals.

To support a comparable assessment of recent trends, the monthly averages for each year were calculated using January through June, matching the months available for 2025. This avoids comparing a partial 2025 total with complete prior years.

Emergency department visits due to opioid poisoning were substantially lower during the first six months of 2025 than during the same months of 2023 and 2024. Hospitalizations showed a similar reduction. Together, these patterns indicate fewer recorded opioid-poisoning encounters requiring emergency or inpatient hospital care during the comparable 2025 period.

The decline is broadly consistent with the reduction observed in paramedic-attended suspected opioid-overdose calls. However, the indicators should not be interpreted as measuring the same events or as proving a common cause. Paramedic responses, emergency department visits, and hospitalizations come from different systems, and some people may reach a hospital without paramedic involvement. Conversely, not every paramedic-attended incident results in an emergency department visit or hospitalization.

Hospitalization should not be described specifically as evidence of a “long-term health effect.” It indicates that the patient was admitted for inpatient care following opioid poisoning, which may reflect the severity of the poisoning, complications, or the need for continued monitoring and treatment.

Age and sex patterns

Adults aged **25-44** recorded the highest number of opioid-poisoning emergency department visits across the selected years. The **45-64** age group ranked second, while substantially fewer visits were recorded among younger and older age groups.

Males recorded considerably more emergency department visits and hospitalizations than females. This indicates that, within the healthcare data used in the dashboard, opioid-poisoning utilization was concentrated among males and adults aged 25-44.

These findings describe the distribution of recorded healthcare encounters, but they do not explain why these groups experienced higher utilization. The data also do not show whether the differences reflect variation in opioid exposure, overdose severity, healthcare-seeking behaviour, population size, or other underlying factors.

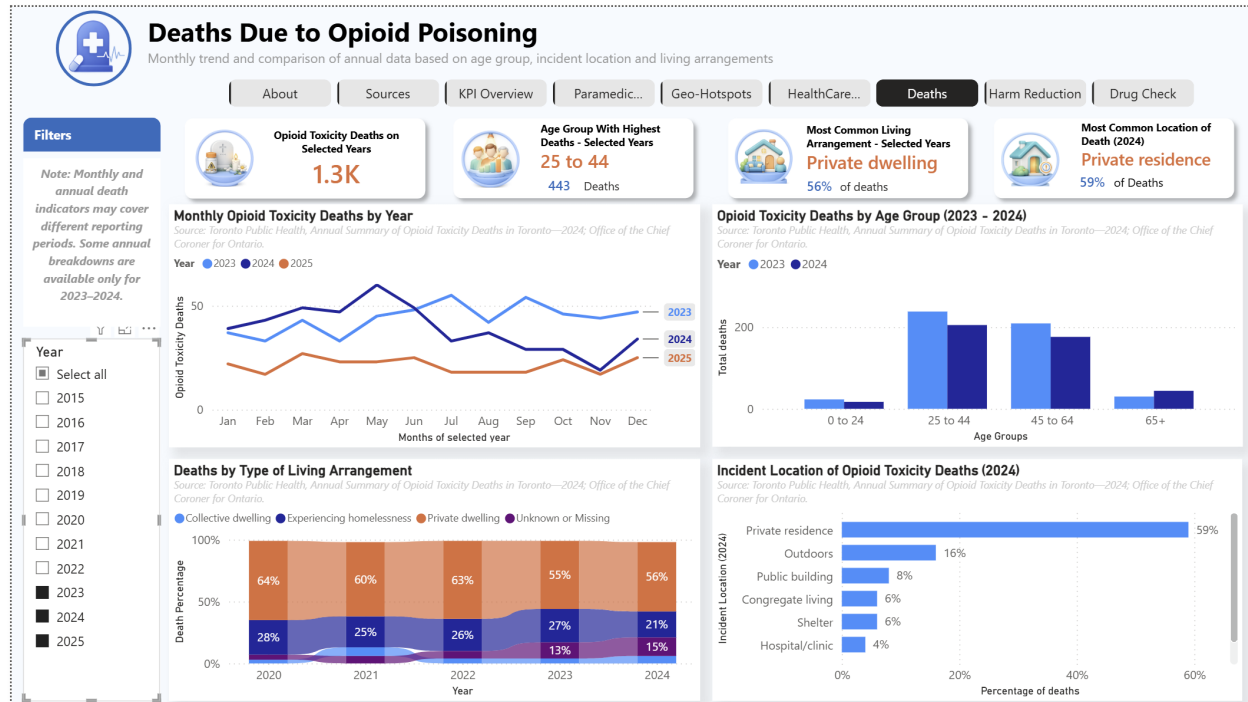
Key finding

Opioid-poisoning emergency department visits and hospitalizations were lower from January to June 2025 than during the same period in 2023 and 2024. The decline was consistent with the downward trend in paramedic-attended suspected opioid-overdose calls, although the indicators measure different healthcare and emergency-service events and do not establish a common cause. Across the demographic data, adults aged 25–44 and males recorded the highest numbers of opioid-poisoning healthcare encounters.

Deaths Due to Opioid Poisoning

Figure 4.

Deaths Due to Opioid Poisoning dashboard page with 2023–2025 selected.



Note. Monthly opioid-toxicity death data and detailed annual breakdowns cover different reporting periods. Age-group data are available for 2023–2024, living-arrangement data are available for 2020–2024, while incident-location data shown in the dashboard refer to 2024.

Monthly opioid-toxicity deaths were generally highest in 2023, declined in 2024, and remained lower throughout 2025. The monthly totals continued to fluctuate, however, indicating that the reduction was not consistent across every month. Toronto Public Health reported 464 confirmed and probable opioid-toxicity deaths among Toronto residents in 2024, representing a decline from recent pandemic-era levels but remaining substantially higher than the pre-pandemic average. (*Toronto Public Health, 2025*)

Age-group pattern

Adults aged **25–44** recorded the highest number of opioid-toxicity deaths across 2023 and 2024, with 443 deaths over the two-year period. The 45–64 age group ranked second. In 2024 alone, adults aged 25–44 represented 46% of accidental opioid-toxicity deaths, compared with 40% among adults aged 45–64. (*Toronto Public Health, 2025*)

This pattern shows that opioid-toxicity mortality was concentrated among working-age adults rather than among the youngest or oldest age groups. However, the age distribution does not explain why these groups experienced more deaths, and age should not be interpreted as a direct cause.

Living arrangement

A **private dwelling** was the most commonly reported living arrangement in every year from 2020 to 2024. In 2024, 56% of accidental opioid-toxicity deaths involved people living in private dwellings. People experiencing homelessness accounted for 21%, while living arrangement was unknown or missing in 15% of cases. (*Toronto Public Health, 2025*)

These findings show that opioid toxicity affects people across different housing circumstances and should not be viewed exclusively as an issue affecting people without stable housing. Living arrangement refers to where the individual was residing at the time of death and may differ from both the location of the poisoning incident and the location of death. (*Toronto Public Health, 2025*)

Incident location

A **private residence** was the most frequently reported location of fatal opioid-toxicity incidents in 2024, accounting for 59% of accidental deaths. Outdoor locations accounted for 16%, while smaller proportions occurred in public buildings, congregate-living settings, shelters, hospitals or clinics, and hotels or motels. (*Toronto Public Health, 2025*)

The predominance of private residences indicates that many fatal incidents occurred in non-public settings. However, the data do not establish whether another person was present, whether the individual was using alone, or how quickly emergency assistance became available.

Possible factors associated with the decline

The decline observed in Toronto is consistent with a broader 17% reduction in opioid-related deaths across Canada from 2023 to 2024. The Public Health Agency of Canada concluded that no single factor explained the national decline and that several factors likely operated together. (Public Health Agency of Canada, 2025).

Changes in the unregulated drug supply were identified as a likely contributing factor, including reductions in fentanyl detections and some high-risk combinations such as opioids mixed with

benzodiazepines. The agency emphasized that the drug supply remained unpredictable and varied across regions. (Public Health Agency of Canada, 2025).

Increased naloxone availability was also identified as a likely factor. Naloxone distribution reached high levels by 2024, and regions distributing more kits experienced larger declines in opioid-related deaths. (Public Health Agency of Canada, 2025).

A declining population at risk of fatal opioid overdose was identified as another likely factor. Possible explanations included the large number of lives lost in previous years, reduced opioid use among youth, and greater declines among some high-risk groups. (Public Health Agency of Canada, 2025).

Changes in consumption behaviours were classified as a possible factor. These included increased smoking rather than injecting, not using drugs alone, and greater use of peer support and other community-led harm-reduction practices. (Public Health Agency of Canada, 2025).

These national findings provide useful context for interpreting Toronto's decline, but they do not establish what caused the local trend. The Public Health Agency of Canada notes that the effects of these factors may differ across regions and that the analysis does not prove cause and effect. (Public Health Agency of Canada, 2025).

Important limitation

The monthly series and detailed annual breakdowns do not necessarily represent identical groups of records. The monthly trend includes confirmed and probable opioid-toxicity deaths across different manners of death, while the detailed age-group, living-arrangement, and incident-location analyses focus primarily on confirmed accidental deaths for which circumstantial information was available. (*Toronto Public Health, 2025*)

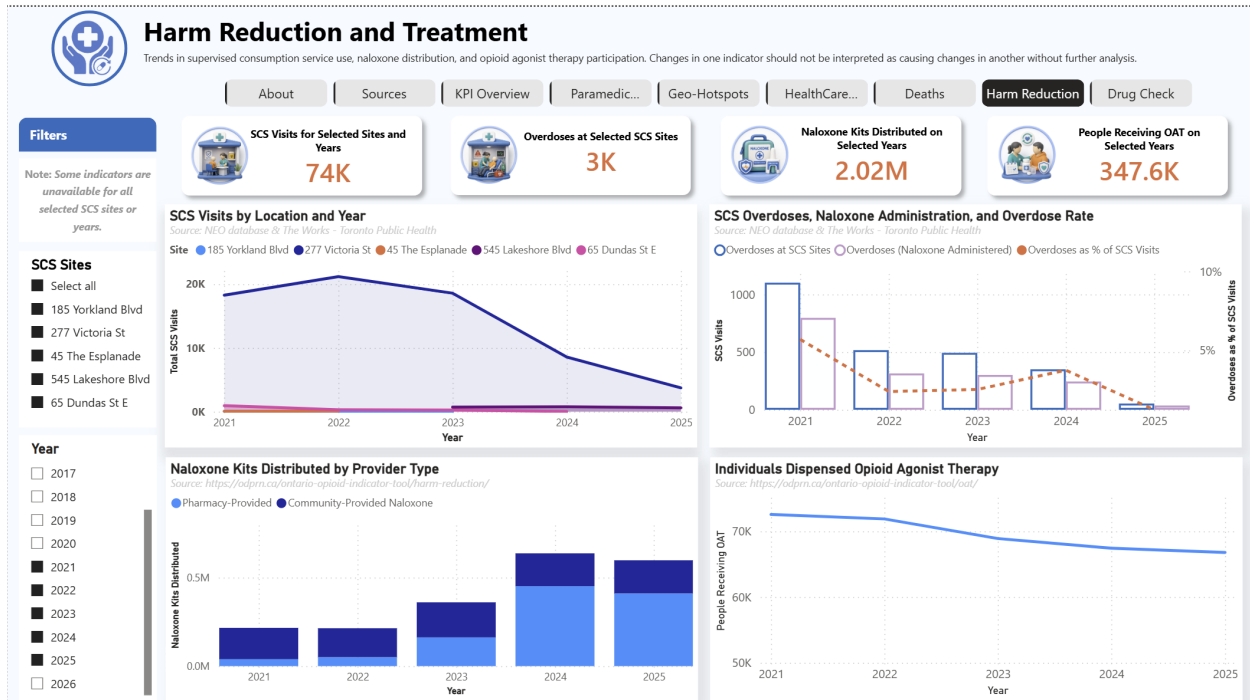
Key finding

Opioid-toxicity deaths declined from the higher levels observed in 2023, but the burden remained substantial. Adults aged 25-44 recorded the highest number of deaths, while private dwellings were the most common living arrangement and private residences were the most frequent incident location. National evidence suggests that changes in the drug supply, naloxone availability, the population at risk, and consumption behaviours may have contributed to the broader decline, although the Toronto data do not establish causation.

Harm Reduction and Treatment

Figure 5.

Harm Reduction and Treatment dashboard page with 2021–2025 selected.



Note. Supervised consumption service data for 2025 are available only through March because Toronto Public Health stopped operating supervised consumption services at 277 Victoria Street after March 31, 2025. Naloxone distribution and opioid agonist therapy data are complete for each calendar year through 2025. The three indicators measure different forms of service activity and should be interpreted separately. Toronto Public Health confirms that it no longer provided supervised consumption services as of April 1, 2025.

Supervised consumption service activity

The Works at **277 Victoria Street** recorded substantially more supervised consumption visits than the other locations shown in the dashboard. However, these sites are not directly comparable. The Works was a long-established, publicly accessible downtown service operating from August 2017 through March 2025. By comparison, the other locations were generally smaller services embedded in temporary shelter settings and operated for shorter periods.

The source notes identify the following operating periods:

- 185 Yorkland Boulevard: March 2021–March 31, 2023

- 45 The Esplanade: June 2021–December 6, 2022
- 545 Lake Shore Boulevard: March 11, 2023–March 31, 2025
- 65 Dundas Street East: December 2020–March 20, 2024
- 277 Victoria Street: August 2017–March 31, 2025

Consequently, differences in total visits likely reflect operating duration, accessibility, service capacity, hours of operation, and the populations served rather than differences in demand alone.

Total SCS visits peaked in 2022 and declined in 2023 and 2024. Although the displayed 2025 total is much lower, it contains only three months of data and must not be compared directly with complete earlier years. The decline before 2025 may also have been affected by site closures, changes in service availability, and reduced operating capacity.

The number of overdose events recorded at the selected sites also declined from the higher level observed in 2021. More importantly, the proportion of SCS visits resulting in an overdose decreased markedly after 2021. Although this proportion increased slightly in 2024, it remained below the 2021 level and declined again in the available January-to-March 2025 data.

These results describe events occurring during service visits rather than unique individuals. One person may account for multiple visits or overdose events. The figures also do not capture overdoses occurring outside the participating Toronto Public Health sites.

Naloxone distribution

Naloxone-kit distribution increased sharply from 2023 to 2024, particularly through pharmacies, and remained high in 2025. This occurred within a broader provincial system that provides free naloxone through participating pharmacies and community organizations. (Government of Ontario, 2023)

Ontario also introduced workplace naloxone requirements on June 1, 2023 for employers meeting specified risk conditions. This policy may have contributed to broader naloxone availability, but the dashboard data do not show how much of the increase was attributable to workplace requirements, pharmacy distribution, community-program expansion, increased demand, repeat distribution, or other changes in program delivery. (Government of Ontario, 2022).

Naloxone-kit counts measure kits distributed rather than unique recipients, overdose events, or successful overdose reversals. A person may receive more than one kit, and the data do not indicate whether a kit was carried, used, replaced, or expired.

Opioid agonist therapy

The number of individuals dispensed opioid agonist therapy declined gradually from 2021 through 2025. Because the decrease occurred across several consecutive years, it represents a sustained downward trend rather than a single-year fluctuation.

The available data describe treatment dispensing but do not explain the reasons for the decline. Potential contributing factors may include changes in treatment retention, clinical practice, patient preferences, access to prescribers or pharmacies, relocation, changes in the population seeking treatment, and the evolving characteristics of the opioid crisis. Additional data would be required to determine whether the trend reflects reduced treatment demand, treatment discontinuation, changes in service delivery, movement to other forms of care, or other factors.

The indicator should be described as **individuals dispensed OAT**, not necessarily people continuously receiving or completing treatment.

Interpretation across indicators

The three measures reflect different aspects of harm reduction and treatment:

- SCS visits measure activity at selected Toronto Public Health service locations.
- Naloxone-kit counts measure distribution through provincial pharmacy and community programs.
- OAT data measure individuals dispensed treatment medication.

The simultaneous decline in SCS visits and OAT dispensing and the increase in naloxone distribution do not show that one service replaced another.

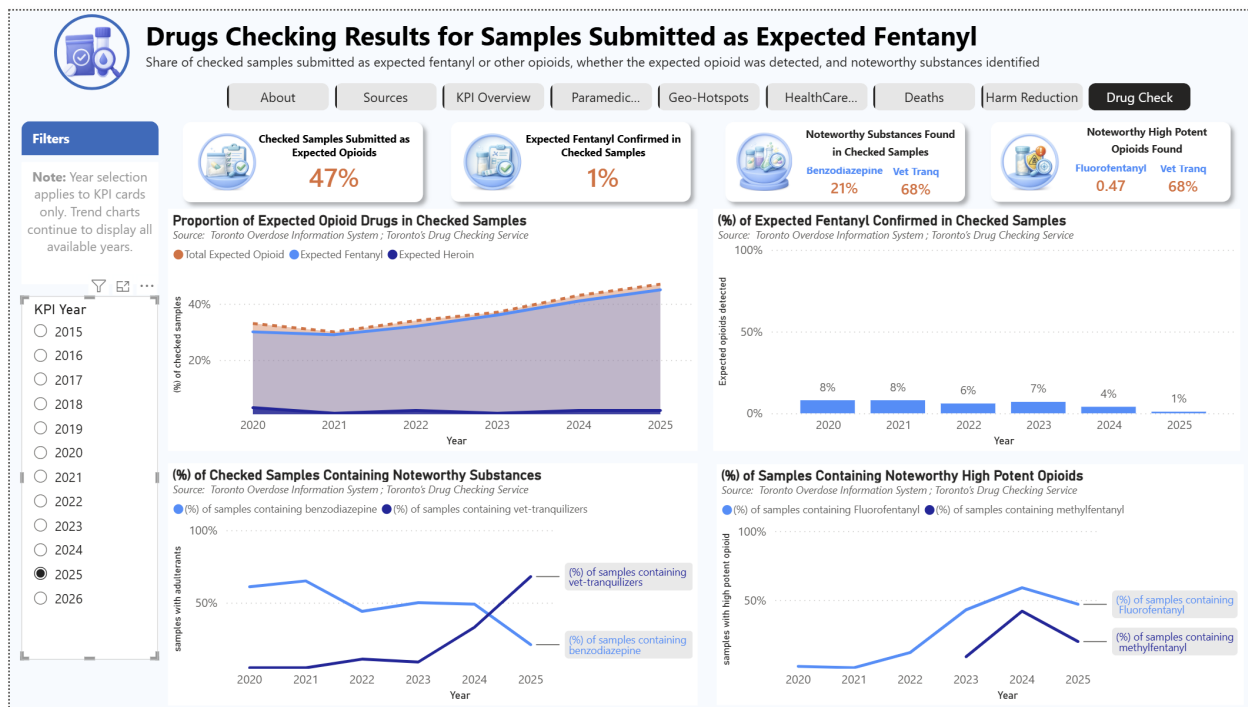
Key finding

Supervised consumption service visits and overdose events declined after 2022, although site closures, different operating periods, and incomplete 2025 SCS data strongly affect the comparison. The proportion of visits involving an overdose remained below its 2021 level. Naloxone-kit distribution rose sharply and remained high through 2025, while the number of individuals dispensed opioid agonist therapy declined gradually. Together, these indicators show changing patterns in service availability, harm-reduction supply, and treatment utilization, but they do not establish causal relationships.

Drug-Checking Results

Figure 6.

Drug-Checking Results for Samples Submitted as Expected Fentanyl dashboard page.



Note. Drug-checking findings represent samples voluntarily submitted to Toronto's Drug Checking Service. They do not represent all drugs circulating in Toronto, the prevalence of opioid use, or the number of people using particular substances. A single sample may contain multiple substances, so detection percentages are not mutually exclusive. The year selector applies only to the KPI cards, while the trend charts display all available years.

Samples submitted as expected opioids

The proportion of checked samples submitted as expected opioids generally increased after 2021, rising from approximately 30% of all checked samples in 2021 to nearly half by 2025. Most of this category consisted of samples submitted as expected fentanyl, while expected heroin represented a comparatively small proportion.

This trend indicates that opioids, particularly expected fentanyl, accounted for an increasing share of the samples brought to the service. It should not be interpreted as evidence that opioid use increased across Toronto because the results reflect only the composition of voluntarily submitted

samples. The increase may instead reflect changes in the substances used by service users, greater concern about the unpredictability of expected-fentanyl samples, or changes in who accessed the drug-checking service.

Whether submitted samples met expectations

Only a small proportion of samples submitted as expected fentanyl contained the expected opioid without the additional unexpected substances captured by this indicator. The proportion declined from approximately 8% in 2020 and 2021 to about 1% in 2025.

This suggests increasing inconsistency between what service users expected and what was detected in the submitted samples. However, the measure must be labelled according to its exact TOIS definition. It should not be described simply as “fentanyl confirmed” when the calculation actually represents samples that contained **only** the expected opioid or otherwise fully met expectations.

The result applies only to voluntarily submitted samples and should not be generalized to Toronto’s entire unregulated drug supply.

Noteworthy substances

The dashboard focuses on the two most frequently detected noteworthy substance categories in samples submitted as expected fentanyl: **benzodiazepine-related drugs** and **veterinary tranquilizers**. Other noteworthy substances were also detected, but their annual detection levels were lower and were not included in the primary trend visual.

Benzodiazepine-related detections were relatively common during the earlier reporting years but declined substantially by 2025. In contrast, detections of veterinary tranquilizers increased sharply in the later years, particularly from 2024 to 2025.

The veterinary-tranquilizer category includes substances such as xylazine and medetomidine. Toronto’s Drug Checking Service first detected xylazine in September 2020 and reports that it is generally found unexpectedly in fentanyl samples. (Toronto’s Drug Checking Service, 2025). Medetomidine was first identified by the service on December 29, 2023; it is described as a veterinary tranquilizer that can be considerably more potent than xylazine. (Toronto’s Drug Checking Service, 2025).

The increase in veterinary-tranquilizer detections suggests a substantial change in the composition of samples submitted as expected fentanyl. Because these substances are often unexpected, their presence may increase uncertainty and health risks for people consuming the

samples. Toronto's Drug Checking Service reported particularly high veterinary-tranquilizer detections in expected-fentanyl samples during early 2025. (Toronto's Drug Checking Service, 2025).

Noteworthy high-potency opioids

The high-potency-opioid visual presents the two most frequently detected selected substances in this category: **fluorofentanyl** and **methyلفentanyl-related drugs**. Other fentanyl analogues and high-potency synthetic opioids were detected during the reporting period but were excluded from the primary chart because their recorded detection levels were lower.

Fluorofentanyl detections generally increased after 2021, reached their highest annual level in 2024, and declined moderately in 2025. Methyلفentanyl-related detections increased sharply in 2024 before declining in 2025.

These substances may be detected alongside fentanyl or in samples where fentanyl itself was not identified. Their presence adds to the variability of expected-fentanyl samples and may increase overdose risk because potency can differ across fentanyl-related compounds.

Toronto's Drug Checking Service notes that high-potency opioids used in combination may increase overdose risk and, in some circumstances, require greater-than-usual naloxone doses. (Toronto's Drug Checking Service, 2025).

Important limitation

The dashboard shows annual percentages rather than the number of tested samples in each category. A high percentage in a year with fewer submitted samples may not represent more detections in absolute terms than a lower percentage in a year with more samples. Changes may also reflect shifts in the types of samples submitted, collection-site activity, testing practices, and service-user participation.

Key finding

Samples submitted as expected opioids, primarily expected fentanyl, accounted for an increasing share of checked samples, while the proportion fully matching expectations declined. The composition of expected-fentanyl samples also changed: benzodiazepine-related detections decreased, veterinary tranquilizers increased sharply, and fluorofentanyl and methyلفentanyl-related detections peaked in 2024 before declining in 2025. These findings demonstrate increasing variability among submitted samples but are not representative of Toronto's entire unregulated drug supply.

Data Notes and Limitations

This dashboard combines indicators from different surveillance, administrative, service, and drug-checking systems. Paramedic calls, emergency department visits, hospitalizations, deaths, supervised consumption service activity, naloxone-kit distribution, opioid agonist therapy, and drug-checking samples measure different events and should not be added together to estimate a total number of overdose incidents or affected individuals.

Reporting periods vary by source. Healthcare-utilization data for 2025 are available only through June, supervised consumption service data only through March, and neighbourhood paramedic data cover 2025 only. Detailed death data also use different periods for age group, living arrangement, and incident location. Partial-year totals should not be compared directly with complete years.

Paramedic data include only suspected opioid-overdose incidents attended by Toronto Paramedic Services. They do not capture all poisonings, and the recorded location may differ from the person's residence or the place where the poisoning began. Neighbourhood rates also depend on 2021 census populations and should be considered alongside total call counts.

Emergency department visits and hospitalizations represent recorded healthcare encounters, not unique individuals. Hospitalizations may follow ED visits, and the data may include people who live outside Toronto but received care in a Toronto hospital. Demographic counts identify groups with the largest recorded utilization but do not necessarily measure population-adjusted risk.

Monthly death trends and detailed annual death characteristics may not represent identical groups of records. Living arrangement and incident location are different measures, and detailed breakdowns may exclude deaths without sufficient circumstantial information.

Supervised consumption service counts represent visits and events rather than unique clients. The sites differed in operating period, capacity, accessibility, and population served, so visit totals are not directly comparable. Site closures also affected annual trends.

Naloxone data measure kits distributed, not unique recipients, use during overdoses, or successful reversals. OAT data measure individuals dispensed treatment medication but do not establish adherence, retention, or treatment outcomes.

Drug-checking findings represent voluntarily submitted samples and are not representative of Toronto's entire unregulated drug supply. A single sample may contain multiple substances, so detection percentages are not mutually exclusive.

The dashboard is descriptive and observational. Similar trends across indicators do not establish that one caused another. External research is used only to provide context, not to prove the causes of Toronto-specific patterns.

Conclusion and Areas for Further Analysis

Conclusion

This dashboard brings together multiple indicators to provide a high-level view of opioid-related harms and responses in Toronto. Paramedic-attended suspected opioid-overdose calls peaked in 2021 and generally declined afterward. Emergency department visits, hospitalizations, and opioid-toxicity deaths also decreased during the most recent comparable reporting periods, although the datasets measure different events and do not establish a shared cause.

The recorded burden was concentrated among males and adults aged 25–44. Geographic analysis also showed a concentration of paramedic-attended incidents in downtown Toronto, with Moss Park recording the highest call count and population-adjusted rate in 2025.

Harm-reduction and treatment indicators followed different patterns. Naloxone-kit distribution increased and remained high, while supervised consumption service visits and the number of individuals dispensed opioid agonist therapy declined. Drug-checking findings continued to show an unpredictable expected-fentanyl supply, including increasing detections of veterinary tranquilizers and the presence of high-potency fentanyl analogues.

Overall, recent declines in several indicators are encouraging, but opioid-related harms remain substantial and unevenly distributed. The findings should be interpreted cautiously because the datasets differ in coverage, definitions, and reporting periods.

Areas for Further Analysis

Future analysis could examine whether the recent declines continue as more complete data become available, particularly for healthcare utilization, deaths, and supervised consumption services. Additional years of neighbourhood-level data would also allow geographic trends to be assessed over time rather than for 2025 alone.

Population-adjusted demographic rates could be used to determine whether the groups with the highest numbers of healthcare encounters and deaths also experience the highest relative burden. Further research could also investigate the decline in OAT dispensing, the effects of changes in service availability, and possible relationships between the evolving unregulated drug supply and overdose-related outcomes.

These areas require more timely, detailed, and consistently defined data and are presented as opportunities for further analysis rather than policy recommendations.

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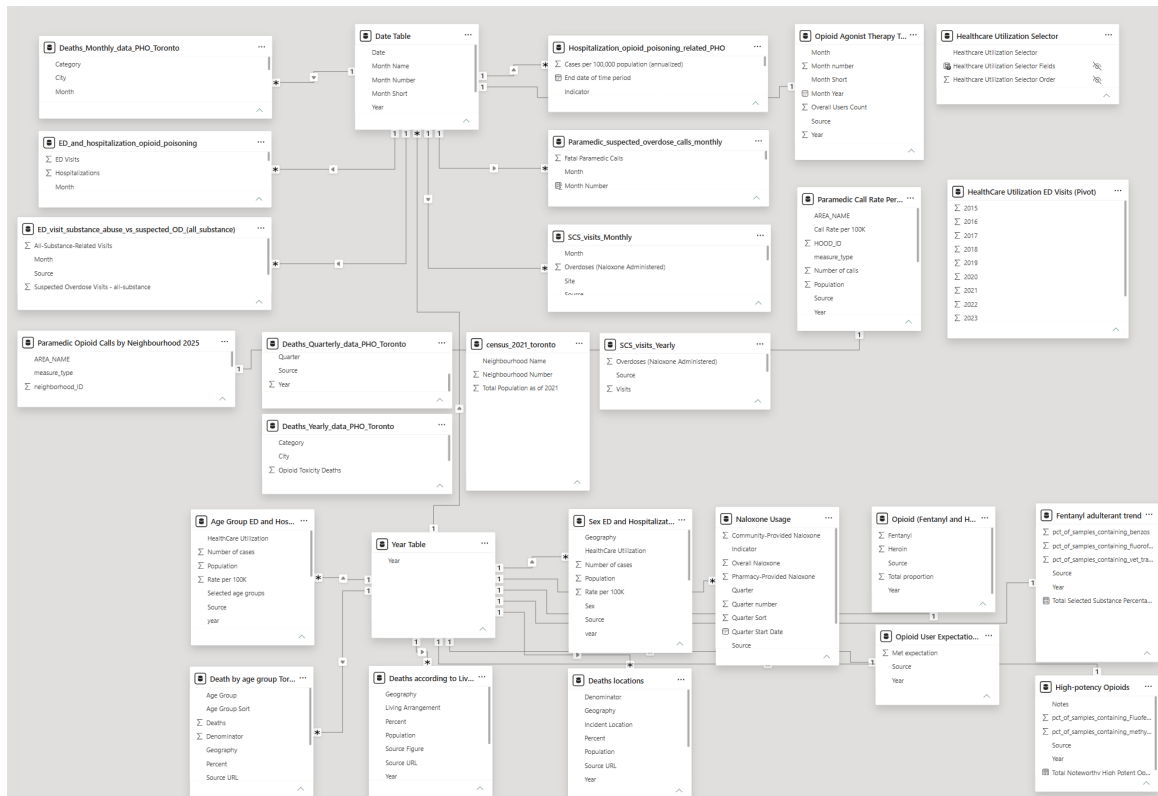
<https://drugchecking.community/report/feb-22-mar-7-2025/>

Appendices

Appendix A. Power BI Data Model

Figure A1

Power BI data model for the Toronto Opioid Overdose Information Dashboard.



Note. Relationships were designed to preserve the different reporting frequencies and units of analysis of each dataset. The indicators were not combined into a single fact table because they represent different events, populations, and reporting systems.

The model integrates multiple subject-specific tables for paramedic responses, healthcare utilization, opioid-toxicity deaths, supervised consumption services, naloxone distribution, opioid agonist therapy, neighbourhood analysis, and drug-checking results. A central Date table supports monthly datasets, while a separate Year table supports annual demographic and circumstantial datasets. The Healthcare Utilization Selector is a disconnected table used to switch selected visuals between emergency department visits and hospitalizations. Some imported or exploratory tables were retained during development but were not used in the final dashboard.